KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY
Deemed to be University
BHUBANESWAR-751024
School of Computer Engineering

Lesson Plan

Artificial Intelligence (CS30002)

Lectures:	3 Hrs / Week	***Internal Assessment Marks:***	*50*
		Activities: *30* ● *Minimum Activities:* ***05*** *and types are quiz, assignment, viva, etc.* ● *Faculty must share the continuous evaluation for* ***15*** *marks each before the Mid-semester and End-semester exams, respectively.* *Mid Sem Exam: 20*	
		End Term Marks:	*50*
Credits:	3		
Groups:	B.Tech.		
Faculty Name:-	Dr. Tanik Saikh		
Contact Details:-	9831149864		

Course Objectives:

CO1:-To understand the various characteristics of Intelligent agents

CO2:-To learn the different search strategies in AI

CO3:-To learn to represent knowledge in solving AI problems .

CO4:-To understand the ways of planning and acting in the real world

CO5:-To know about the models behind the AI application.

Module No. & Name	Topic/Coverage	No. of lectures	Lecture Serial no.
1. Introduction	1.Introduction:- Use and Application. 2. Definition:- Rationality, Thinking Humanly, Acting Humanly, Thinking Rationally and Acting Rationally. Turing Test, Four Capabilities for A.I system. 3.Future of Artificial Intelligence.	2	1-2
2. Intelligent Agents	1.Characteristics of Intelligent Agents:- Agent Autonomy, Actuators ,Sensors, Environment, Performance Measure, Agent function and Agent Program. (Vacuum Cleaner Example, etc.) 2. Agents and Environment:- Rational Agent , Discuss various environments, Specification of Task Environment (Using Examples). 3. Typical Intelligent Agents and their Types:- Simple Reflex, Model based, Goal based and Utility based.(Discuss with Diagram).	3	3-6
3. Solving Problems by Searching	1.Defining a problem for state space searching. (State Space Representation of Water-Jug Problem, N-Queen Problem , Monks and Demons problem, 8-Puzzle problem, etc.) *(One or Two problem to be explained in class others can be given for practice).*	1	7-19
	2. Search Strategies:- Search Tree, Solution Path, Nodes,Open List, Closed List, concept of space and time complexity.	**1**	

	3.Uninformed Strategies:- BFS , Uniform Cost Search , DFS , Iterative Deepening, Depth Limited and Bidirectional. Discuss the Space and Time complexity of each Strategy.	**8**	
	4. Informed (Heuristics Strategies) :- Concept of Heuristics , Admissibility and consistency, Greedy Best First Search , A* Algorithm. Discuss Admissibility, Consistency and Optimality of A*.	**3**	
4. Beyond Classical Search	1.Local Search Algorithms and Optimization Problems :- Objective Function , Global and Local Minimum/Maximum , Hill Climbing , Problems with Hill Climbing and Solution, Steepest Hill Climbing , Simulated Annealing, Genetic Algorithm (Fitness Function, Crossover and Mutation).	4	20-27
	2. Backtracking Search:- Concept of Constraint Satisfaction Problem , Formulation of problem into CSP. (Crypt-Arithmetic Problem and Map Coloring Problem).	**2**	
	3. Adversarial Searching :- Concept of Two Players Game, Min-Max Algorithm , Alpha-Beta Pruning. (Tic-tac-toe as an Example)	**2**	
5. Knowledge Representation.	1. Basic of Proposition Logic , Truth Tables, Atomic Sentences, Complex Sentences, Quantifiers , Connectives. 2. First Order Predicate Logic. 3. Unification. 4. Forward Chaining and Backward Chaining.	5	28-32

	5. Resolution. 6. Knowledge Representation using First order Predicate logic. 7. Logical Agents (Knowledge-based agents, the Wumpus World, entailment, inference, sound and complete inference algorithms, propositional logic, various inference procedures such as model checking and theorem proving, forward and backward chaining etc.)		
6.Planning	1. Planning with state-space search. 2. Partial-order planning. 3. Planning graphs, 4. planning and acting in the real world. 5. Plan generation systems.	2	33-35
7.Probabilistic Reasoning.	1. Uncertainty and Review of probability. 2.Probabilistic Reasoning. 3.Bayesian networks. 4.Inferences in Bayesian networks. 5. Temporal models and Hidden Markov models.	2	36-37
	END SEMESTER EXAM		

Text Books:

1. Artificial Intelligence: A Modern Approach, Stuart Russel, Peter Norvig, Pearson Education

Reference Books:

1. Artificial Intelligence, Rich, Knight and Nair, Tata McGraw Hill.
2. Principles of Artificial Intelligence, Nils J. Nilsson, Elsevier, 1980.

Evaluation Scheme:

ES No.	Evaluation Component	Percentage of Evaluation
1	Mid-Semester Examination	20
2	Activities	30
3	End-Semester Examination	50

Activity based Teaching and Learning:

Considering the guidelines circulated and after discussing with the faculty members, following activity based teaching and learning is proposed:

- **Activity List**

  Component wise distributions of the activities are listed below.

  - Problem Solving Assignment
  - Critical Thinking Assignment
  - Quiz
  - Viva / Presentation

Course Materials: Course Material will be provided for all topics which can be used as reference. The material consists of –

- Lecture Notes
- Class Work
- Home Work
- Supplementary Reading (including online study aids)

Machine Learning (CS31002)
Lesson Plan

Course: CS31002 (Machine Learning)
Credits: 4
Session: December-2024 to May-2025
Coordinator: Dr. Partha Pratim Sarangi (pp.sarangifcs@kiit.ac.in)
Co-coordinator: Dr. Santos Kumar Baliarsingh (santos.baliarsinghfcs@kiit.ac.in)

Course Objective

1. To provide a broad survey of different machine-learning approaches and techniques
2. To understand the principles and concepts of machine learning
3. To learn regression and classification models
4. To learn different clustering models
5. To understand artificial neural networks (ANN) and Convolutional neural networks (CNN) concepts
6. To develop programming skills that help to build real-world applications based on machine learning

Course Outcomes

Upon completion of the course, the students will be able to:

CO1: Solve typical machine learning problems

CO2: Compare and contrast different data representations to facilitate learning

CO3: Apply the concept of regression methods, classification methods, and clustering methods.

CO4: Suggest supervised /unsupervised machine learning approaches for any application

CO5: Implement algorithms using machine learning tools

CO6: Design and implement various machine learning algorithms in a range of real-world applications.

Total Lectures ≈ 48
Before Mid-Sem ≈ 24
After Mid-Sem ≈ 24

Module 1

Lecture	Topics
1	Introduction to Machine Learning, definition, and real-world applications.
2	Types of machine learning - Supervised, Unsupervised, Semi supervised learning, Definitions and examples.
3	Regression - Linear Regression, Intuition, Cost Function
4	Linear Regression - Gradient Descent
5	Multiple Linear regression
6	Closed-form Equation, Type of Gradient Descent (Batch, Stochastic, Mini-batch) - Definition, properties.
7	Normalization and Standardization (definition and why), Overfitting and Underfitting
8	Bias, Variance, Bias and Variance tradeoff
9	Regularization - Lasso Regularization, Ridge Regularization
10	ACTIVITY-1

Module 2

Lecture	Topics
11	Classification, Logistic Regression - 1 (binary)
12	Logistic Regression - 2 (binary)
13	Nearest neighbor and K Nearest Neighbour

14	Error Analysis - Train/Test Split, validation set, Accuracy, Precision, Recall, F-measure, ROC curve, Confusion Matrix
15	Naive Bayes Classifier - 1
16	Naive Bayes Classifier - 2
17	Decision Tree: Introduction, Id3 Algorithm - 1
18	Decision Tree - Id3 Algorithm - 2
19	Decision Tree - Problem of Overfitting, Pre-pruning/post-pruning Decision Tree, Examples.
20	Support Vector Machine - Terminologies, Intuition, Learning, Derivation - 1
21	Support Vector Machine - Terminologies, Intuition, Learning, Derivation - 2
22	Support Vector Machine - KKT Condition - 3
23	Support Vector Machine - Kernel, Nonlinear Classification, and multi-class (Basic concept) - 4
24	ACTIVITY-2

Mid Semester

25	Principal Component Analysis - Steps, merits, demerits, Intuition - 1
26	Principal Component Analysis - Steps, merits, demerits, Intuition - 2
27	Understanding and Implementing PCA using SVD for dimensionality reduction

Module 3

Lecture	Topics
28	Clustering: Introduction, K-means Clustering - 1
29	K-Median Clustering - 2
30	K-Means Clustering – 3 (Numerical)
31	DBSCAN Clustering - Why we use?, parameters, characterization of points, steps, determining parameters, time/space complexities
32	Mean Shift Clustering
33	Hierarchical Clustering - Agglomerative Clustering, Single/Complete/Average/Centroid Linkage
34	Hierarchical Clustering - Divisive hierarchical clustering
35	ACTIVITY-3

Module 4

Lecture	Topics
36	Introduction Neural networks, McCulloch-Pitts Neuron
37	Least Mean Square (LMS) Algorithm
38	Perceptron Model
39	Multilayer Perceptron (MLP) and Hidden layer representation
40	Non-linear problem solving, Activation Functions
41	Backpropagation Algorithm - 1
42	Backpropagation Algorithm - 2
43	Exploding Gradient Problem and Vanishing Gradient Problem, why and how to avoid
44	Introduction to Convolutional Neural Network (CNN)
45	Basic idea about their working and structure
46	Data Augmentation, Batch Normalization, Dropout
47	ACTIVITY-4

Module 5

Lecture	Topics
48	Introduce machine learning tools like Scikit Learn, PyTorch, TensorFlow, Kaggle competitions, etc.
	Case Study (Any Two)
Case Study – 1: Classification using Iris Dataset Case Study – 2: Feature Extraction using PCA for Wine Dataset Case Study – 3: Implement linear regression to predict house prices based on features like size, location, and number of rooms. Case Study – 4: Clustering using Iris Dataset Case Study – 5: Classification of MNIST Dataset using CNN Model	

Activities

Task	Marks
Before Mid-semester	
Quiz - 1	15 marks reduced to 5
Survey - 1	5
Coding - 1	5
After Mid-semester	
Quiz - 2	15 marks reduced to 5
Survey - 2	5
Coding - 2	15

Textbooks:

1. Madan Gopal, “Applied Machine Learning”, TMH Publication
2. Kevin P. Murphy, “Probabilistic Machine Learning”, MIT Press, 2023.
3. Ethem Alpaydin, “Introduction to Machine Learning”, Fourth Edition, MIT Press, 2010.

Reference Books:

1. Laurene Fausett, “Fundamentals of Neural Networks, Architectures, Algorithms and Applications”, PearsonEducation, 2008.
2. C. M. Bishop, “Pattern Recognition and Machine Learning”, Springer, 2007.
3. Simon Haykin, “Neural Networks and Learning Machines”, Pearson 2008

Course Title: Software Project Management
Course Code: CS30012
(L-T-P-Cr: 3-0-0-3)
Prerequisites: Software Engineering (CS31001)

Faculty
Prof. Kunal Anand
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Contact No: +91-8464800019
Chamber: F14, Block-A; Campus-15
Visiting Hours: Monday to Friday (6 PM - 6:30 PM)

Course Objectives:

1. Recognize basic concepts and issues of software project management
2. Emphasize successful software projects that support organization's strategic goals
3. Comprehend software quality issues
4. Comprehend software risk issues
5. Analyse SPM tools

Course Contents:

UNIT I : SPM Concepts:
Definition, Components of SPM, Challenges and opportunities, Tools and techniques, Managing human resource and technical resource, Costing and pricing of projects, Training and development, Project management techniques.*

UNIT II : Software Measurements:
Monitoring & measurement of SW development, Cost, Size and time metrics, Methods and tools for metrics, Issues of metrics in multiple projects.*

UNIT III : Software Quality:
Quality in SW development, Quality assurance, Quality standards and certifications, The process and issues in obtaining certifications, The benefits and implications for the organization and its customers, Change management.*

UNIT IV :Risk Issues:
The risk issues in SW development and implementation, Identification of risks, Resolving and avoiding risks, Tools and methods for identifying risk management.*

UNIT V : SPM Tools:
Software project management using Primavera & Redmine, Case study on SPM tools.* *Programming assignments are mandatory.

Course Outcomes:
Upon completion of this course, the students will be able to:
CO1: Identify the job roles of an IT project manager to conduct project planning activities
CO2: Plan to maintain and monitor software projects and processes
CO3: Design and develop project modules and assign resources
CO4: Comprehend, assess, and estimate the cost of risk involved in a project management
CO5: Analyze the tools for risk management
CO6: Design a Case study using SPM tools

Day-wise Plan

Lecture:	3 Hrs/Week	**Internal Assessment Marks:**	50
Tutorial:	0 Hrs/Week	**End Term Marks:**	50
Practical:	0	**Credits:**	3

Topics / Coverage	**Unit-wise CO mapping**	**Lecture No**
Why is software project management important?, What is a project?, Software projects versus other types of project, Definition of SPM	**UNIT I : SPM Concepts** **CO: 1** **CO: 2** **CO: 3**	1
Components of SPM: Activities covered by software project management Plans, Tools and techniques: methods and methodologies		2
Categorization of software projects, Challenges and opportunities: Project success and failure, What is management? Management control.		3
Managing human resource and technical resource: Project Portfolio Management, Costing and pricing of projects: Cost-benefit Evaluation, Risk evaluation		4
Managing Allocation of Resources, Benefits Management, Training and development		5
An Overview of Project Planning/Project management techniques: Introduction to Step Wise project Planning: Step 0: Select project, Step 1: Identify project scope and objectives, Step 2: Identify project infrastructure Step-3: Analysis of project Characteristics,		6,7,8
Step 4: Identify project products and activities Step 5: Estimate effort for each activity, Step 6: Identify activity risks,		9,10,11
Step 7: Allocate resources Step 8: Review/publicize plan, Steps 9 and 10: Execute plan/lower levels of planning		12,13, 14
Monitoring & measurement of SW development: Software Effort Estimation: Introduction, Where are estimates done?, Problems with over- and underestimates, The basis for software estimating	**UNIT II : Software Measurements**	15
Methods and tools for Cost, Size and time metrics: Software effort estimation techniques, Bottom-up estimating, The top-down approach and parametric models		16
Expert judgment, Estimating by analogy, Albrecht function point Analysis, Function points Mark II, COSMIC full function points		17, 18
COCOMO, Issues of metrics in multiple projects		19, 20
Pre-Mid semester Session (04/12/2024 – 15/02/2025)		

Mid Semester Examination (17/02/2025 – 22/02/2025) **Mid Sem Syllabus (Unit 1 and Unit 2)**		
Post-Mid semester Session (23/02/2025 – 11/04/2025)		
Quality in SW development, The place of software quality in project planning, Quality assurance, Quality standards and certifications, The process and issues in obtaining certifications	**UNIT III : Software Quality**	21
The importance of software quality: The benefits and implications for the organization and its customers, Defining software quality, ISO 9126, Product versus process quality.		22, 23
Process capability models, SEI-CMM, Techniques to enhance software quality, Testing Quality plans, Change management		24,25
Risk, Categories of risk, A framework for dealing with risk, Steps in risk management	**UNIT IV :Risk Issues** **CO: 4** **CO: 5**	26
Risk identification, Risk assessment, Risk planning, Risk management		27
Evaluating risks to the schedule, Applying the PERT technique, Monte Carlo simulation, Critical chain concepts		28,29
Software project management using Primavera & Redmine	**UNIT V : SPM Tools** **CO: 6**	30,32
Case study on SPM tools		33-36
End Semester Examination (12/04/2025 – 22/04/2025)		

Textbooks:

1. Sanjay Mohapatra, "Software Project Management", Cengage, 2024

2. Richard H. Thayer, "Software Engineering Project Management", Second Edition, John Wiley & Sons,2001.

3. Royce, Walker, "Software Project Management", First Edition, Pearson Education, 1998.

Reference Books:

1. Kelker S. A., "Software Project Management", Third Edition, PHI, 2003

2. Kan, Stephen H., "Metrics and Models in Software Quality Engineering", Addison-Wesley Longman Publishing Co. Inc., 2002.

3. Galin, Daniel, "Software Quality Assurance: From Theory to Implementation", Addison-Wesley,2004.

Internal Assessment : Activity Based Continuous Evaluation (30 Marks) + Mid Semester (20 Marks)

Activity 1	Pre-Midsem (15 Marks)	15 Marks should be communicated to the students before Mid Semester Starts.	All Activities should be some kind of analytical exercise/test. The minimum number of activities should
Activity 2			

Activity 3	Post-Midsem (15 Marks)	30 Marks should be communicated to the students before Mid Semester Starts.	be 4, However, Faculty members can conduct more than 4 Activities.
Activity 4			

SCHOOL OF CIVIL ENGINEERING
KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY
DEEMED TO BE UNIVERSITY, BHUBANESWAR - 24

Date: 15.02.2025
Spring Semester 2024-25
Course Handout

1.	**Course code**	**:**	CE 30072
2.	**Course Title**	**:**	Fundamentals of Project Management
3.	**Credit**	**:**	03
4.	**Pre-requisite**	**:**	Nil
5.	**Course Faculty** **Course Coordinator**	**:** **:**	Prof. (Dr.) P.K. Acharya Prof. R.Panda
6.	**Course Description**	**:**	This course is designed to enable the students to understand and appreciate the importance of concepts of project management. The students would be able to investigate complex business problems and propose project-based solutions
7.	**Course Outcomes**	**:**	After successful completion the course, the students will be able to **CO 1:** Appreciate the importance of various aspects of project management, **CO 2:** Apply knowledge of various domains of the project to address specific management needs, **CO 3:** Understand the concepts related to time management of the project, **CO 4:** Learn the aspects related to the optimization of project time and cost, **CO 5:** Understand the facts concerning resource management, and **CO 6:** Learn about the various concepts related to cost management and engineering economy.
8	**Course Content:**		
	Introduction to project management: Introduction to project management, Purpose of project management, Process of project management, Objectives of project management, Elements of a network diagram, Rules of a network diagram, Constraints of a network diagram, Errors in a network diagram. **Performance domains of the project:** Different performance domains of the project – Stakeholders, Team, Development approach and life cycle, Planning, Project work, Delivery, Measurement, Uncertainty. **Time management of the project:** Critical path method (CPM) – Critical path analysis, Activity times and floats, Project evaluation review technique (PERT) – Three times estimates, Beta-distribution curve, Critical path analysis for PERT network, Probability of completion of the project, Differences between CPM and PERT. **Optimization of time and cost of the project:**		

	Various costs associated with the project, Variation of various costs of the project concerning the time of completion, Cost slope, and Optimization of the mathematical model of the network. **Resources management of the project:** Resource allocation, Resource smoothening, and Resource leveling. **Finance management of the project:** Principles of engineering economy, Interest, and interest formulae, Comparison of alternatives, Minimum cost point analysis, Breakeven point analysis, Depreciation, and Depletion.
9.	**Text and Reference Books:** 1. A Guide to The Project Management Body of Knowledge (Pmbok® Guide), Project Management Institute, 7th Edition, 2021, ISBN 9781628256642. 2. U.K. Shrivastav, Construction Planning and Management, Galgotia Publications Pvt. Ltd, 3rd Edition, 2005, Reprint 2015, ISBN-978-81-7515-246-5. 3. C.F. Gray and E.W. Larson, Project Management, the Managerial Process, McGraw-Hill, 6th Edition, 2017, ISBN-13 - 978-9339212032. 4. K.N. Jha, Construction Project Management, Pearson Education, 2nd Edition, 2015, ISBN-10 – 9332542015.

10	**Lesson Plan**	**Date**	
Lecture Number	**Learning topics to be covered**		
1	Introduction to the subject	18.12.2024 19.12.24	CO1
2	Introduction to project management, Purpose of project management,	03.01.2025	
3	Process of project management, Objectives of project management,	08.01.2025	
4	Elements of a network diagram, Rules of a network diagram, Constraints of a network diagram, Errors in a network diagram.	10.01.2025	
5	Activity-1 (Date: 15.01.2025)		
6	Performance domains of the project- Stakeholders, Team,	17.01.2025	CO2
7	Performance domains of the project- Development approach and life cycle, Planning, Project work	22.01.2025	
8	Performance domains of the project- Delivery, Measurement, Uncertainty.	24.01.2025	
9	Activity-2 (Date: 29.01.2025)		
10	Time management of the project- Critical path method (CPM) – Critical path analysis, Activity times, and floats.	31.01.2025	CO3

11	Time management of the project- Project evaluation review technique (PERT) – Three times estimates, Beta-distribution curve	05.02.2025	
12	Time management of the project- Critical path analysis for PERT network, Probability of completion of the project, Differences between CPM and PERT.	07.02.2025	
13	Doubt clearing/ Revision class	12.02.2025	
14	Activity-3 (Date: 14.02.2025)		
	Mid-Semester Examination (17.02.2025 to 22.02.2025)		
15	Optimization of time and cost of the project	28.02.2025	CO4
16	Various costs associated with the project, Variation of various costs of the project concerning the time of completion, Cost slope,	07.03.2025	
17	Optimization of the mathematical model of the network.	12.03.2025	
18	Activity- 4 (Date:14.03.2025)		
19	Resources management of the project- Resource allocation	19.03.2025	CO5
20	Resources management of the project- Resource smoothening	21.03.2025	
21	Resources management of the project- Resource leveling	26.03.2025	
22	Activity- 5 (Date:28.03.2025)		
23	Finance management of the project- Principles of Engineering Economy,	02.04.2025	CO6
24	Finance management of the project- Interest and interest formulae, Comparison of alternatives,	04.04.2025	
25	Finance management of the project- Minimum cost point analysis, Breakeven point analysis, Depreciation, Depletion	09.04.2025	
26	Activity- 6 (Date:11.04.2024)		
End Semester Examination (12.04.2025 to 22.04.2025)			

*The details of the activities will be notified prior to the conduct of the activity in class.

11. Evaluation

Sl. No.	Evaluation Component	Marks
1	Mid Semester Examination	20
2	End Semester Examination	50
3	Active Learning Activities: Critical Thinking Focus/ Creation / Interactivity Focus / Quiz / Reflection	30

12. Attendance: Every student is expected to be regular (in attendance) in all lecture classes, tutorials, labs, tests, quizzes, seminars, etc., and in fulfilling all tasks assigned to him/her. Attendance will be recorded and 75% attendance is compulsory.

13. Makeup Examination:

1) No make-up examination will be scheduled for the mid-semester examination. However, official permission to take a make-up examination will be given under exceptional circumstances such as admission to a hospital due to illness/injury, or calamity in the family at the time of examination.

2) A student who misses a mid-semester examination because of extenuating circumstances such as admission to a hospital due to illness/injury, or calamity in the family may apply in writing via an application form with supporting document(s) and medical certificate to the Dean of the School for a make-up examination.

3) Applications should be made within five working days after the missed examination.

Course Co-ordinator
Prof P.K. Acharya

UNIVERSAL HUMAN VALUES (HS30401)
Lesson Plan

COURSE OBJECTIVE

The objective of the course is to develop a holistic perspective based on self-exploration, understand the harmony in the human being, strengthen self-reflection, and develop commitment and courage to act.

COURSE OUTCOMES

After successfully completing the course, the students will be able to

CO1: Understand the concept of value education and its need,
CO2: Apply their knowledge on value education for apt self-assessment,
CO3: Comprehend human-human relationship,
CO4: Build holistic perception of harmony at all levels of existence,
CO5: Develop the sense of natural acceptance of human values, and
CO6: Create people friendly and eco-friendly environment.

Topic to be cover

SL. No	Topic	No. of Lecture
Module 1:	**Need, Basic Guidelines, Content and Process for Value Education.**	
1	Purpose and motivation for the course, recapitulation from UHV-I A look at basic Human Aspirations	1
2	Continuous Happiness and Prosperity	1
3	Right understanding, Relationship and Physical Facility	1
4	The basic requirements for fulfilment of aspirations of every human being with their correct priority	1
5	Understanding Happiness and Prosperity correctly- A critical appraisal of the current scenario	1
6	Course Introduction - Need, Basic Guidelines, Content and Process for Value Education.	1
7	Self-Exploration–what is it?	1
8	Self-Exploration–Its content and process	1
9	‘Natural Acceptance’ and Experiential Validation- as the process for self-exploration	1
10	Method to fulfil the above human aspirations: understanding and living in harmony at various levels.	1
	Activity 1	
Module 2:	**Understanding Harmony in the Human Being - Harmony in Myself!**	
11	Understanding human being as a co-existence of the sentient ‘I’ and the material ‘Body’	**1**
12	Understanding the needs of Self (‘I’) and ‘Body’ - happiness and physical facility	**1**

13	Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer)	**1**
14	Understanding the characteristics and activities of 'I' and harmony in 'I'	**1**
15	Understanding the harmony of I with the Body: Sanyam and Health; correct appraisal of Physical needs, meaning of Prosperity in detail	**1**
16	Programs to ensure Sanyam and Health.	**1**
	Activity 2	
Module 3:	**Understanding Harmony in the Family and Society- Harmony in Human-Human Relationship**	
17	Understanding values in human- human relationship	**1**
18	Human- human relationship: meaning of Justice (nine universal values in relationships)	**1**
19	program for its fulfilment to ensure mutual happiness; Trust and Respect as the foundational values of relationship	**1**
20	Understanding the meaning of Trust; Difference between intention and competence	**1**
	Activity 3	
21	Understanding the meaning of Respect,	**1**
22	Difference between respect and differentiation	**1**
23	the other salient values in relationship i.e.: Affection, Care, Guidance, Reverence, Glory, Gratitude, and Love	**1**
Mid-Semester Examination: 20th February 2025 - 26th February 2025		
24	Understanding the harmony in the society (society being an extension of family):	**1**
25	Resolution, Prosperity, fearlessness (trust) and co-existence as comprehensive Human Goals	**1**
26	Visualizing a universal harmonious order in society- Undivided Society, Universal Order- from family to world family.	**1**
Module 4:	**Understanding Harmony in the Nature and Existence - Whole existence as Coexistence**	
27	Understanding the harmony in the Nature	**1**
28	Relationship of Mutual Fulfillment	**1**
29	Interconnectedness and mutual fulfilment among the four orders of nature- recyclability and self-regulation in nature	**1**
30	Understanding Existence as Co- existence of mutually interacting units in all-pervasive space	**1**
31	Holistic perception of harmony at all levels of existence.	**1**
	Activity 4	
Module 5:	**Implications of the above Holistic Understanding of Harmony on Professional Ethics**	
32	Natural acceptance of human values; Definitiveness of Ethical Human Conduct	**1**
33	Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order	**1**
34	Competence in professional ethics: a. Ability to utilize the professional competence for augmenting universal human order b. Ability to identify the scope and characteristics of people- friendly and eco-friendly production systems, c. Ability to identify and develop appropriate technologies and	**1**

	management patterns for above production systems.	
	Activity 5	
35	Case studies of typical holistic technologies, management models and production systems	**1**
36	Strategy for transition from the present state to Universal Human Order: a. At the level of individual: as socially and ecologically responsible engineers, technologists and managers b. At the level of society: as mutually enriching institutions and organizations	**1**
	Activity 6: Quiz test	
End Semester Examination: 12th April 2025 - 22nd April 2025		

Text Book

1.Human Values and Professional Ethics by R R Gaur, R Sangal, G P Bagaria, Excel Books, New Delhi, 2010

Reference Books

1. Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.
2. , A.N. Tripathi, Human ValuesNew Age Intl. Publishers, New Delhi, 2004.
3. The Story of Stuff (Book).

Lesson Plan:

Applications Development Laboratory (**Machine Learning Applications**)

Course Code CS33002 (L-T-P-Cr: 0-0-4-2)
Credits: 2
Session: December-2024 to May-2025
Course faculty: Prof. Ipsita Paul

Week 1: Introduction to Machine Learning

Objective: Understand the basics of machine learning and Python / R,etc

Experiment:

- Write a Python program to demonstrate data preprocessing steps: handling missing values, encoding categorical data, and feature scaling.
- Load a sample dataset using pandas (e.g., Iris or a custom dataset).
- Plot the distribution of a feature using matplotlib.pyplot.hist().
- Create scatter plots to understand relationships between features using seaborn.scatterplot().
- Use a correlation heatmap to find the relationship between multiple features with seaborn.heatmap().

Week 2: Regression Analysis

Objective: Learn simple linear regression for predictive analysis.

Experiment:

Implement simple linear regression to predict house prices based on features such as area and number of rooms.

- Evaluate the model using metrics such as mean_squared_error and r2_score.

- Visualize the regression line on a scatter plot of the data

Note: *A common dataset we can explore is the "House Prices - Advanced Regression Techniques" dataset from kaggle. The UCI Machine Learning Repository has many datasets for classification, regression, and clustering tasks. A popular dataset related to housing is the "California Housing Prices" dataset.*

Week 3: Multiple Linear Regression

Objective: Extend regression analysis to handle multiple predictors.

Experiment:

Implement multiple linear regression to predict student performance based on study hours, class attendance, and assignment scores.

- Calculate metrics like Mean Squared Error (MSE) and R^2 Score to assess performance.

- Create a scatter plot comparing the actual vs. predicted values

Note: We can get Student Performance Dataset from kaggle

Week 4: Classification with Logistic Regression

Objective:Understand binary classification using logistic regression.

Experiment:

Build a logistic regression model to classify emails as spam or not spam.

- Accuracy: A percentage indicating how many emails were correctly classified.

- Assess model performance using accuracy, confusion matrix, and classification report (Classification Report: Precision, Recall, and F1-score for both spam and non-spam classes)

Note: We will use a publicly available dataset such as the SpamBase dataset from the UCI Machine Learning Repository or create a simple synthetic dataset for demonstration purposes

Week 5: Decision Trees

Objective: Learn decision tree classifiers for non-linear decision boundaries.

Experiment:

Develop a decision tree model to classify species in the ***Iris dataset.***

- Evaluate the model using accuracy and a detailed classification report with precision, recall, and F1-score for each species.

- Use plot_tree() to visualize the decision tree. This helps understand how the model splits the data at different nodes.

Week 6: Random Forests

Objective: Understand ensemble methods using random forests.

Experiment:

Implement a random forest classifier to predict customer churn in a telecom dataset.

- Accuracy: Find overall prediction accuracy.

- Confusion Matrix: Breakdown of true/false positives and negatives.

- Classification Report: Includes precision, recall, and F1-score for both churned and retained classes.

Note: You can use a real-world dataset such as the Telco Customer Churn Dataset available on Kaggle or create a similar dataset for practice.

Week 7: Support Vector Machines (SVM)

Objective: Explore SVM for linear and non-linear classification.

Experiment:

Build an SVM model to classify handwritten digits using the MNIST dataset.

- Accuracy: Overall percentage of correct predictions.

- Confusion Matrix: Provides a breakdown of correct/incorrect predictions for each digit.

- Classification Report: Detailed precision, recall, and F1-score for each digit.

Note:The MNIST dataset consists of 70,000 grayscale images of handwritten digits (28x28 pixels) representing numbers from 0 to 9.
We can use sklearn.datasets.fetch_openml to load the dataset.

Week 8: K-Means Clustering

Objective: Learn unsupervised learning through clustering.

Experiment:

Apply K-Means clustering to group customers based on purchasing patterns.

- Plot the clusters in 2D space (using scaled features) and mark the centroids.

- Analyze each cluster to identify customer groups based on purchasing patterns, such as:

- High-income, high-spending customers.
- Low-income, low-spending customers

Note: We can use the Mall Customers Dataset. This dataset is available in .csv format, typically containing columns such as CustomerID, Gender, Age, Annual Income , and Spending Score (1-100).

Week 9: Principal Component Analysis (PCA)

Objective: Reduce dimensionality using PCA.

Experiment:

Perform PCA on a high-dimensional dataset and visualize the results.

- Displays the proportion of variance captured by each principal component.
- 2D Plot: Projects data onto the first two principal components.
- 3D Plot: Projects data onto the first three principal components.

Note: We can use the Wine Dataset from the sklearn library. It has 13 features (high dimensionality) related to chemical properties of wine samples, which can be reduced to 2 or 3 principal components for visualization.

Week 10: Neural Networks with TensorFlow/Keras

Objective: Introduce deep learning basics with neural networks.

Experiment:

Build a simple neural network to classify handwritten digits using TensorFlow/Keras.

- The images are flattened from 28x28 matrices into 1D vectors of length 784 to be fed into the neural network.
- Input Layer: The input is a flattened vector of size 784 (28x28).
- Hidden Layer: A dense layer with 128 neurons and ReLU activation. This layer allows the model to learn complex patterns.

- Dropout Layer: A dropout layer with a rate of 0.2, which helps reduce overfitting by randomly setting some of the weights to zero during training.

- Output Layer: A softmax layer with 10 neurons for classifying the digits from 0 to 9.

- Show the Test Accuracy, model's performance over epochs(accuracy plot), loss plot indicates whether the model is converging

Note: We can use MNIST Dataset: The MNIST dataset contains 60,000 training images and 10,000 test images of handwritten digits (0-9).

Week 11: Convolutional Neural Networks (CNN)

Objective: Learn image classification using CNNs.

Experiment:

Implement a CNN model to classify images from the CIFAR-10 dataset.

- Show the Test Accuracy, Training vs Validation Accuracy, Training vs Validation Loss and Predictions

Note: CIFAR-10 Dataset: The CIFAR-10 dataset contains 60,000 32x32 color images in 10 classes. CIFAR-10 dataset can be loaded using cifar10.load_data().

Capstone Application (Any of the application must be deployed on the web and accessible through a browser**)**

Objective: Integrate learned concepts into a real-world application.

Experiment:

Hosting an ML application on the web with a front end that can be run in a browser including creating the machine learning model, developing a web interface (front end), setting up a back-end server, and hosting the entire application on the web.

Create a Web Front-End (User Interface):

Languages: Use HTML, CSS, and JavaScript (or frameworks like React, Angular, or Vue.js) to create the user interface.

Develop the Back-End (Server):

Flask or FastAPI (for Python): These are lightweight web frameworks for serving machine learning models.

Connect the Front-End to the Back-End:

Use AJAX or Fetch API in JavaScript to send user input to the back-end (Flask/FastAPI)

Learning Outcomes:

- Understand and apply various supervised and unsupervised machine learning algorithms.
- Build and evaluate models using Python and its libraries (e.g., NumPy, pandas, scikit-learn, TensorFlow).
- Solve real-world problems using machine learning techniques.

Tools Required:

- Python 3.x/ R
- Libraries: NumPy, pandas, matplotlib, scikit-learn, TensorFlow/Keras, NLTK/Spacy

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY
Deemed to be University
BHUBANESWAR-751024

Spring Semester 2024

Course Handout

1. **Course code** : CS 39002
2. **Course Title** : **Artificial Intelligence Laboratory (AI Lab.)**
3. **LTP Structure** :

L	T	P	Total	Credit
2	0	0	2	1

4. **Course Faculty** :
   Name : Dr. Tanik Saikh
   Email ID : tanik.saikhfcs@kiit.ac.in
   Faculty Room : FB-402, Fourth Floor, Campus-14

5. **Course offered to the School :** School Computer Engineering

6. **Course Objective:**
   - To provide skills for designing and analyzing AI based algorithms.
   - To enable students to work on various AI tools.
   - To provide skills to work towards solution of real life problems

General Instructions:
Install Python and use the VS Code IDE to implement the assignments given for AI Lab. or,

Use Google Colab to implement the assignments given for the AI Lab. Students must use their KIIT mail ID to create Google Colab Notebooks. Naming convention can be "RollNo_AssignmentNo.ipynb."

Provide a brief overview of searching techniques, adverserial search in the context of artificial intelligence.

Assignment-1: Maze Solver using BFS and DFS

Objective: Implement BFS and DFS to solve a maze.

Problem Statement: Given a grid-based maze where 0 represents walls and 1 represents walkable paths, find the shortest path from a start cell to an end cell.

Tasks:
- Use BFS to find the shortest path.
- Use DFS to explore all possible paths and report one valid path (not necessarily the shortest).
- Compare the number of nodes explored by BFS and DFS.

Assignment 2: Route Finder Using Bi-Directional BFS/DFS

Objective: Use Bi-directional BFS/DFS to solve a navigation problem.

Problem Statement: Represent a city map as a graph where intersections are nodes and roads are edges. Find the shortest path between two locations.

Tasks:
- Implement Bi-directional BFS to minimize the number of nodes explored.
- Compare the performance of Bi-directional BFS with standard BFS and DFS.
- Visualize the search process (e.g., using a library like networkx in Python).

Assignment 3: Search for Treasure using the Best-First Search

Objective: Use Best-First Search to find a treasure in a grid.

Problem Statement: The treasure is hidden in a grid, and each cell has a heuristic value representing its "closeness" to the treasure. Implement Best-First Search to locate the treasure.

Tasks:
- Use Manhattan distance as a heuristic.
- Implement the algorithm to always move to the most promising cell first (minimum heuristic value).
- Analyze how heuristic choice affects performance.

Assignment 4: Uniform Cost Search for Optimal Path

Objective: Implement Uniform Cost Search for a weighted graph.

Problem Statement: Given a weighted graph (e.g., a transportation network with travel costs), find the minimum-cost path between two nodes.

Tasks:
- Represent the graph as an adjacency list.
- Implement Uniform Cost Search to find the optimal path.
- Compare it with BFS for unweighted graphs.

Assignment 5: A Search for a Puzzle Solver*

Objective: Solve the 8-puzzle using A* search.

Problem Statement: The 8-puzzle involves sliding tiles to achieve a goal state. Use A* to solve it.

Tasks:

Define heuristic functions:

- H1: Number of misplaced tiles.
- H2: Sum of Manhattan distances of all tiles from their goal positions.
- Implement A* with both heuristics.
- Compare the performance of the two heuristics in terms of the number of nodes explored and solution depth.

Assignment 6: Path Planning for a Robot

Objective: Use A* Search to find an optimal path for a robot navigating a 2D grid.

Problem Statement: A robot must move from a start point to a goal in a grid while avoiding obstacles.

Tasks:

Implement A* with:

- The Manhattan distance heuristic applies to grids without any diagonal movement.
- The Euclidean distance heuristic is applicable to grids that allow diagonal movement.
- Use a plotting library to visualize the found path.
- Compare A* with BFS and Uniform Cost Search.

Assignment 7: Game AI Using Search Algorithms

Objective: Implement AI to solve a simple turn-based game.

Problem Statement: Design an AI agent to play a game (e.g., Tic-Tac-Toe or Snake and Ladder) using search algorithms.

Tasks:

- Use BFS and DFS for exploring game states.
- Implement A* Search with a heuristic function to improve efficiency.
- Compare search strategies for different game board configurations.

Assignment 8: Navigation with Multiple Goals

Objective: Solve a problem where multiple goals exist using search algorithms.

Problem Statement: A robot in a grid needs to collect items (goals) before reaching an exit. Each goal has a different priority or cost.

Tasks:
- Use BFS/DFS for simpler scenarios (unweighted goals).
- Implement A* or Uniform Cost Search for weighted scenarios.
- Analyze the trade-offs between path length and goal priority.

Assignment 9: AI Planner Using A for Task Scheduling*

Objective: Use A* Search to optimize task scheduling.

Problem Statement: A set of tasks with dependencies and durations needs to be scheduled to minimize total time.

Tasks:
- Represent tasks and dependencies as a directed graph.
- Use A* Search when the heuristic estimates the remaining tasks' duration.
- Compare results with a greedy algorithm.

Assignment 10: Comparative Study

Objective: Evaluate and compare different search algorithms.

Problem Statement: Given a domain (e.g., pathfinding, puzzle solving), evaluate BFS, DFS, Bi-directional BFS, Uniform Cost Search, Best-First Search, and A* Search.

Tasks:
Analyze:
- Efficiency: Nodes explored, time taken.
- Optimality: Whether the solution is optimal.
- Create visualizations to compare algorithms

Assignment 11: Implementation of local optimization techniques, such as Hill Climbing, for solving AI-based search problems.

Assignment 12: Implementation of Genetic Algorithm (GA) for solving AI-based search problems.

Assignment 13: Solving the Water-Jugs Problem Using Adversarial Search

Develop a competitive version of the Water-Jugs problem where two players take turns making moves. Implement the Minimax algorithm to determine the optimal strategy for the AI player. Extend the solution by incorporating alpha-beta pruning for improved efficiency.

Problem Setup:
You are given two jugs of capacities A liters and B liters, and a target volume T liters. Two players alternately make moves, with the objective of the game being to reach

exactly T liters in any jug. The first player to achieve this wins. If no valid moves are left, the game ends in a draw.

Rules for Valid Moves:

1. Fill either jug completely.
2. Empty either jug completely.
3. Pour water from one jug into the other until one is empty or the other is full.

Game Design:

1. Implement the Water-Jugs problem as a two-player game.
2. Represent the game state as a tuple (x,y)(x, y)(x,y), where xxx and yyy are the current volumes in the two jugs.

Minimax Implementation:

1. Use the Minimax algorithm to evaluate all possible moves for both players.
2. Define a utility function:

   - Positive scores for states closer to the target volume TTT for the AI player.
   - Negative scores for states closer to the target for the opponent.
   - Zero for states with no valid moves left.

Alpha-Beta Pruning Integration:

1. Enhance the Minimax implementation by adding alpha-beta pruning to improve efficiency.

Player Interaction:

1. Allow a human player to compete against the AI agent.
2. Alternate turns between the human player and the AI.

Testing and Analysis:

1. Test the AI's performance against a human opponent and analyze its strategies.
2. Compare execution times for the Minimax algorithm with and without alpha-beta pruning.

Deliverables:

1. A Python program implementing the game with:

   - Minimax algorithm.
   - Alpha-beta pruning.

2. A brief report explaining:

- The rules and structure of the game.
- Design choices for the utility function.
- Results of testing the AI's performance.

Assignment 14: Assignment: Solving the 4-Queens Problem Using Adversarial Search

Design a competitive version of the 4-Queens problem where two players alternately place queens on a 4x4 chessboard. Implement the Minimax algorithm to strategize queen placements and incorporate alpha-beta pruning for efficient decision-making.

Problem Setup:
The game involves two players (AI and Human) alternately placing queens on a 4x4 chessboard. The goal is to place queens such that no two queens threaten each other. The player unable to make a valid move loses the game.

Rules for Valid Moves:

1. A queen can be placed in any cell of the chessboard, provided it is not attacked by any previously placed queen.
2. A queen attacks cells in the same row, column, and diagonals.
3. Players take turns placing one queen at a time.
4. The game ends when no valid moves are left or all 4 queens are placed.

Game Design:

1. Represent the chessboard as a 4x4 grid.
2. Design functions to check valid placements and update the board.

Minimax Implementation:

1. Use the Minimax algorithm to evaluate all possible moves for both players.
2. Define a utility function:
    1. Positive scores for AI-favorable positions.
    2. Negative scores for opponent-favorable positions.
    3. Zero for neutral or draw positions.

Alpha-Beta Pruning Integration:

1. Optimize the Minimax implementation with alpha-beta pruning.

Player Interaction:

1. Alternate turns between the human player and the AI agent.
2. Display the board after each move for visualization.

Testing and Analysis:

1. Test the AI's performance against a human opponent.
2. Compare execution times for the Minimax algorithm with and without alpha-beta pruning.

Deliverables:

1. A Python program implementing the 4-Queens game with:
    1. Minimax algorithm.
    2. Alpha-beta pruning.
2. A brief report explaining:
    1. The rules and structure of the game.
    2. Utility function design.
    3. Results of testing the AI's performance.

Assignment 15: Solving the Tower of Hanoi Problem Using Adversarial Search

Transform the traditional Tower of Hanoi puzzle into a competitive two-player game. Implement adversarial search strategies using the Minimax algorithm to solve the problem. Enhance efficiency by incorporating alpha-beta pruning.

Problem Setup:

The Tower of Hanoi involves three rods and a set of N disks of decreasing size stacked on the first rod. The goal is to move all the disks to the last rod following these rules:

1. Only one disk can be moved at a time.
2. A disk can only be placed on top of a larger disk or an empty rod.
3. Moves alternate between two players (AI and Human).

The player who moves the largest disk to the target rod wins. If no valid moves are left and the game hasn't ended, it is considered a draw.

Game Design:

1. Represent the rods and disks using lists (e.g., rods = [[3, 2, 1], [], []] for N=3N=3N=3).
2. Create a function to validate moves and update the rods.

Adversarial Search:

1. Implement the Minimax algorithm to evaluate all possible moves for both players.
2. Define a utility function:

    - Assign positive scores for moves favorable to the AI (e.g., moving larger disks closer to the target rod).
    - Assign negative scores for moves favorable to the opponent.

Alpha-Beta Pruning:

1. Integrate alpha-beta pruning to optimize the search process by pruning unnecessary branches.

Player Interaction:

1. Alternate turns between the AI and the human player.
2. Display the state of the rods after each move.

End Condition:

1. The game ends when:

   - All disks are moved to the target rod.
   - No valid moves are left.

2. The winner is determined based on the size of the largest disk moved to the target rod by each player.

Deliverables:

1. A Python program implementing:

   - The Tower of Hanoi game mechanics.
   - Minimax algorithm.
   - Alpha-beta pruning optimization.

2. A brief report explaining:

   - The rules and structure of the game.
   - Utility function design.
   - Results of testing the AI's performance.

Assignment 16: Intelligent Solving of the 8-Puzzle Problem Using Heuristic Search

Implement and compare heuristic search strategies for solving the 8-Puzzle problem. Specifically, develop solutions using the **A*** algorithm and **Greedy Best-First Search (GBFS)**, leveraging heuristics like the Manhattan Distance or the number of misplaced tiles.

Problem Setup:

The **8-Puzzle** is a sliding puzzle consisting of a 3x3 grid with numbered tiles (1-8) and a blank space. The goal is to arrange the tiles in a specified order (usually numerical) by sliding tiles into the blank space, starting from a given configuration.

Representation of the Problem:

1. Model the puzzle as a 3x3 grid using a 2D list or a flat list of 9 elements.

2. Define the possible moves: Up, Down, Left, Right.
3. Represent the goal state as [1, 2, 3, 4, 5, 6, 7, 8, 0] (where 0 represents the blank space).

Heuristic Functions:

1. Implement heuristic functions to guide the search:
    1. **Manhattan Distance**: The sum of the absolute differences between the current and target positions of each tile.
    2. **Misplaced Tiles**: The number of tiles not in their correct positions.

Search Algorithms:

1. **A***: Combines the cost to reach a state (g(n)) with the heuristic estimate to the goal h(n).
2. **Greedy Best-First Search (GBFS)**: Considers only the heuristic value (h(n)) for each state.

Implementation Requirements:

1. Write a function to expand possible moves from a given state.
2. Implement a priority queue to explore states based on the chosen heuristic.
3. Ensure the solution handles unsolvable configurations gracefully.

Performance Evaluation:

1. Compare the performance of A* and GBFS in terms of:

    - Number of nodes expanded.
    - Solution optimality (path cost).
    - Execution time.

2. Test with multiple initial configurations.

Deliverables:

1. A Python program that:

    - Implements the 8-Puzzle problem with A* and GBFS.
    - Supports both heuristic functions.

2. A brief report discussing:

    - The implementation of heuristics.
    - Comparison of algorithms based on performance metrics.
    - Observations and conclusions.

Total Assignments: 16

Faculties are free to increase the number of assignments based on time and students' interest. However, the given assignments must not be reduced further (that is, less than 16).

Evaluation:

Aim is to examine student's understanding in problem solving using suitable AI concepts and implementation of the solution using Python.

Marks Distributions[$]:

1.	Continuous Evaluation (daily basis)	: 20 Marks
2.	Lab Test-I (pre-midterm) and Lab Test-II (post-midterm)	: 15 Marks
3.	Quiz	: 10 Marks
4.	Viva	: 05 Marks
5.	Lab record hard-copy (hand written complete code)	: 20 Marks
6.	Sessional Test (External)	: 30 Marks

[$]*There is a flexibility to change the marks ditribution based on requirements. However, theese components should be part of the overall 100 marks computation.*

References:

1. How to use Google Colab for Python: https://github.com/cserajdeep/Google-Colab-Helper
2. Python Coding Tutorial-1: https://www.youtube.com/watch?v=nLRL_NcnK-4
3. Python Coding Tutorial-1: https://www.youtube.com/watch?v=rfscVS0vtbw